# PSEUDO-COHERENCE IS UNCHANGED BY ADJOINING A VARIABLE AND KILLING IT

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

*Editorial note.* The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

*Editorial note.* The counted proof span contains the author's remark that the result is also a special case and that a second proof follows, together with the \medskip\noindent separator; all of it is delivered verbatim. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

*Allowed standard prerequisites (editorial).*

- Pseudo-coherence is preserved by pushforward along a ring map. invoked at proof lines 21268-21269 and 21274-21276. Gives one implication and supplies the special-case observation.
- Pseudo-coherence is preserved by the derived pullback of complexes and the pull-push comparison. invoked at proof lines 21278-21284. Produces the direct sum K^bullet oplus K^bullet[1] over R.
- A direct summand of a pseudo-coherent complex is pseudo-coherent. invoked at proof lines 21286-21287. Removes the extra shifted copy and finishes the second implication.

*Source reference.* Original lemma 16283: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16283–16283. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

*Source reference.* Original lemma 16345: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16345–16345. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

*Source reference.* Original lemma 21218: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21218–21218. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 16196: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16196–16196. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

**Lemma 0.2.** *Let $R$ be a ring and $K^\bullet$ a complex of $R$-modules. Let $m \in \mathbf{Z}$. Consider the $R$-algebra map $R[x] \to R$ which maps $x$ to zero. Then $K^\bullet$ is $m$-pseudo-coherent as a complex of $R$-modules if and only if $K^\bullet$ is $m$-pseudo-coherent as a complex of $R[x]$-modules.*

*Proof.* This is a special case of Lemma 16283. We also prove it in another way as follows.

Note that $0 \to R[x] \to R[x] \to R \to 0$ is exact. Hence $R$ is pseudo-coherent as an $R[x]$-module. Thus one implication of the lemma follows from Lemma 16283. To prove the other implication, assume that $K^\bullet$ is $m$-pseudo-coherent as a complex of $R[x]$-modules. By Lemma 16345 we see that $K^\bullet \otimes^{\mathbf{L}}_{R[x]} R$ is $m$-pseudo-coherent as a complex of $R$-modules. By Lemma 21218 we see that $K^\bullet \oplus K^\bullet[1]$ is $m$-pseudo-coherent as a complex of $R$-modules. Finally, we conclude that $K^\bullet$ is $m$-pseudo-coherent as a complex of $R$-modules from Lemma 16196. $\square$